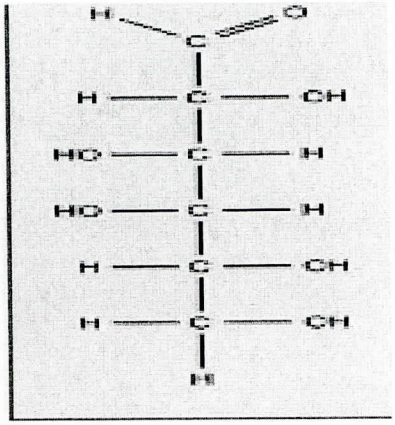
Biochemistry Cume – February 9, 2018

Total points: 100

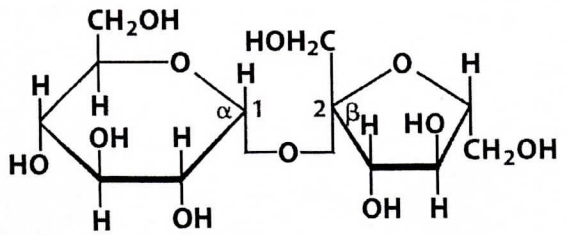
1. Given below is the linear structure of galactose



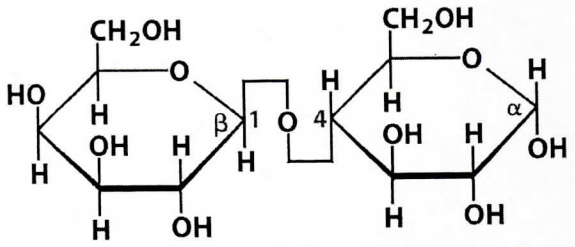
Define the following types of stereoisomers. Draw the chemical structures and identify the stereoisomers of galactose that are relevant to the individual definition. **Total points 24 (6 + 6 + 6 + 6) points**

- **A.** Enantiomers
- **B.** Diastereomers
- **C.** Epimers
- **D.** Anomers

2. Given below are the chemical structures of the disaccharides, lactose and sucrose. Explain why lactose is a reducing sugar and sucrose is not. **20 points**



Sucrose



Lactose

3. A bamboo stem is found to grow at a phenomenal rate of 0.3 meters/day. Given that bamboo stems are composed entirely of cellulose fibers oriented in the direction of growth, calculate the number of glucose residues per second that must be added enzymatically to growing cellulose units to account for the growth rate. Each glucose unit in the cellulose molecule is about 0.45 nm long. (The molecular mass of a glucose unit in glycogen is 162 Daltons (**20 points)**

4. 25 mg sample of E.coli glycogen was hydrolyzed in 2 mL of 2N sulfuric acid. The hydrolysate was neutralized and diluted to 10 mL. The glucose content of the final solution was found to be 2.35 mg/mL. Calculate the purity of the isolated glycogen, (The molecular mass of a glucose unit in glycogen is 162 Daltons) **36 points**